

Geogrid Embankment (with slip)

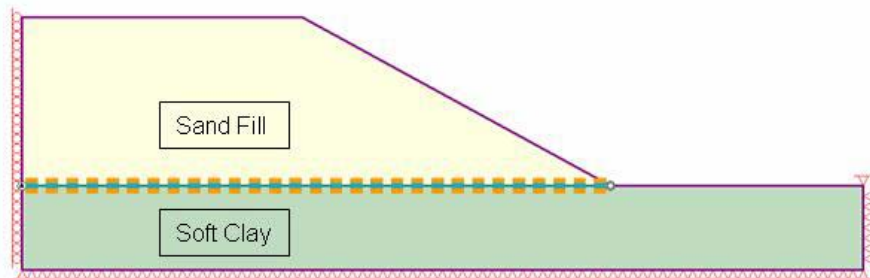
In Tutorial 11, we performed a shear strength reduction (SSR) analysis for a sand embankment on top of a soft clay layer with a geogrid liner in between. We will now proceed with the same model and add the following features:

1. Allow slip between the geogrid liner and the soil layers.
2. Construct the model in two stages. The first stage simulates only the clay layer. The geogrid and the sand fill embankment are added in the second stage.

Topics Covered

- Shear strength reduction
- Slope stability
- Staging
- Multiple materials
- Initial element loading
- Structural interfaces
- Joints
- Liner support (geogrids)

Geometry

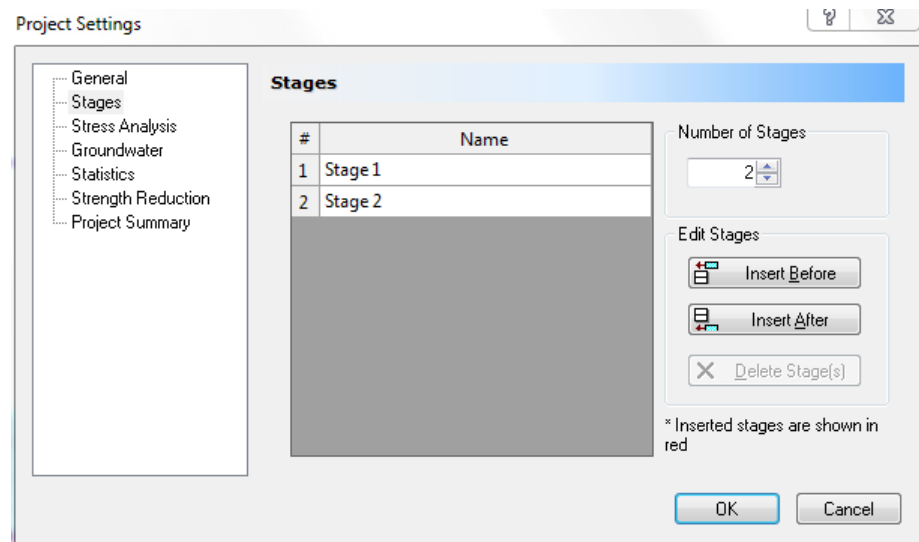


Model

Start the *Phase2* Model program and open the .fez file you created in Tutorial 11. If you did not do Tutorial 11 then open the file **Tutorial 11 Geogrid Embankment (no slip).fez** from the Examples > Tutorials folder of your *Phase2* installation folder.

Project Settings

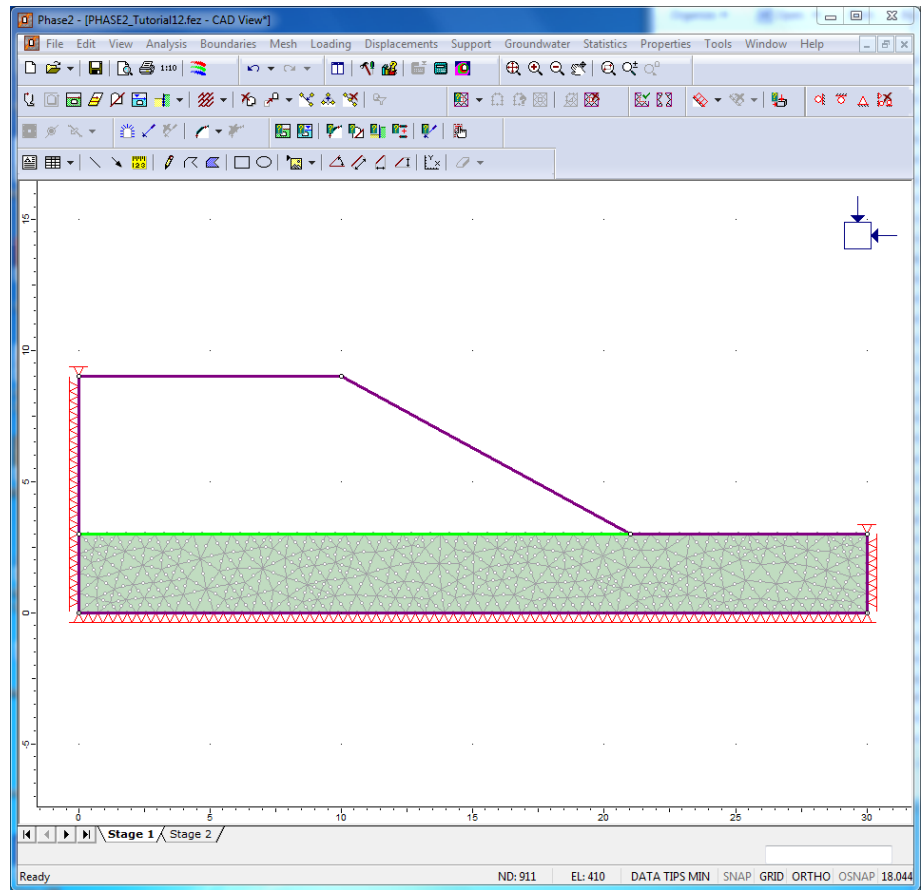
Open the Project Settings dialog from the Analysis menu and make sure the General tab is selected. Change the number of stages from 1 to 2. Click OK to save and close the dialog.



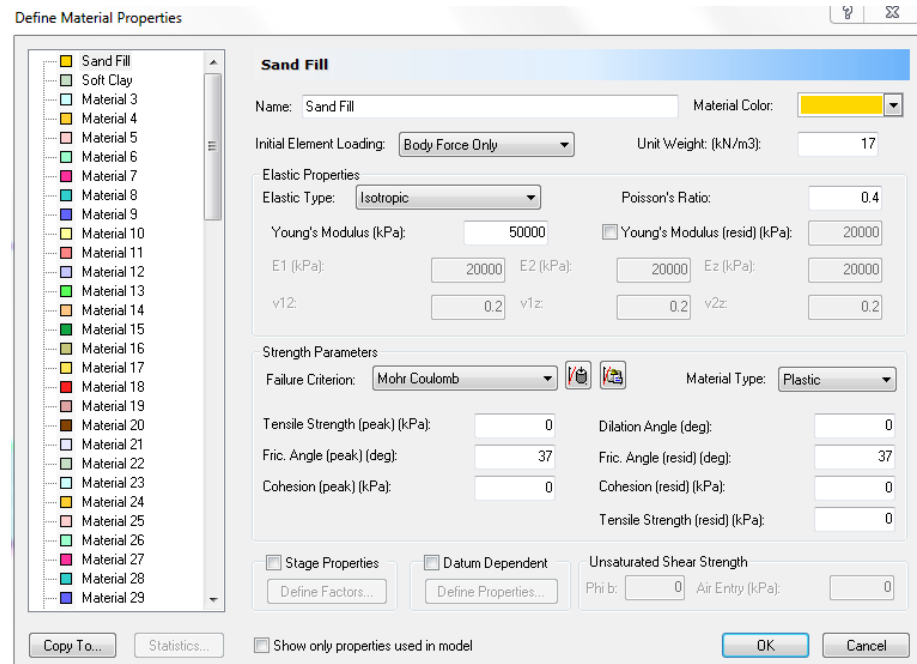
Staging

In this tutorial we wish to build the model in a more realistic way than in Tutorial 11. The first stage will consist of only the clay layer. In the second stage we will then add the geogrid and the sand fill embankment. We do this by applying the following steps:

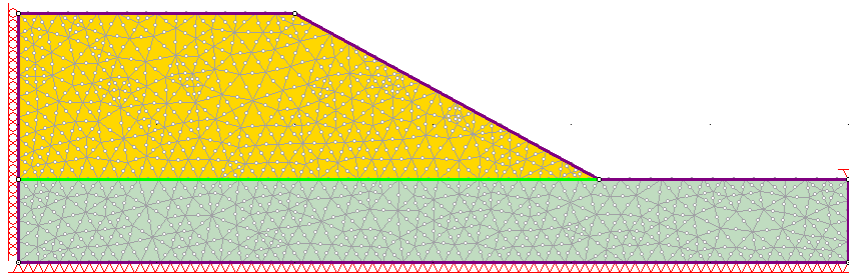
1. Click on the Stage 1 tab at the bottom of the window. Right click inside the Sand Fill layer and choose Assign Material. A list of possible materials will now appear. Choose Excavate. This will delete the sand fill layer.
2. Right click on the geogrid liner. Choose Delete Liner from the popup menu. Your model should now appear as in the following figure.



3. Click on the Stage 2 tab. Add the sand fill embankment by right clicking in the excavated zone, choosing Assign Material, and then choosing Sand Fill.
4. Since the sand fill is manually deposited on top of the existing clay layer, the sand fill will not have an in-situ stress associated with it. The stress that develops in the sand fill will be purely due to its own weight. Therefore we need to change the sand fill properties. Right click inside the sand fill. Select Material Properties. For Initial Element Loading select Body Force Only as shown. For more information on initial element loading consult the *Phase2* Help.



- The sand fill will likely settle under its own weight. To enable this settlement, we need to allow the left boundary to move in the vertical direction. To do this, right click on the left boundary of the sand embankment. Select Restrain X. The clay layer will also compact under the weight of the sand so right click on the left boundary of the clay layer and select Restrain X. Finally, we want the bottom left corner to be restrained both in X and Y, therefore right click on the bottom boundary and select Restrain X,Y. The model should now look like this.



The model will now simulate the deposition of a sand fill embankment in the second stage.

Geogrid

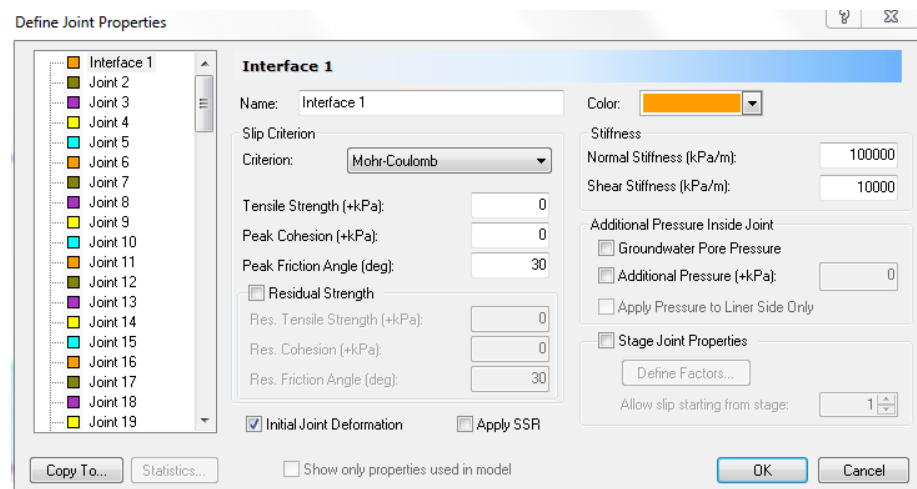
We now wish to add a geogrid between the clay layer and the sand fill in Stage 2 of the simulation. In Tutorial 11, we modelled the geogrid as a simple liner. This is a good method if there is no slip between the reinforcement layer and the soil. To simulate a geosynthetic with slip, we need to create a Structural Interface.

The Structural Interface option in *Phase2* allows you to model support which has a sliding interface (joint) on BOTH sides of the support element. We therefore need to assign Joint properties as well as Liner properties.

Properties

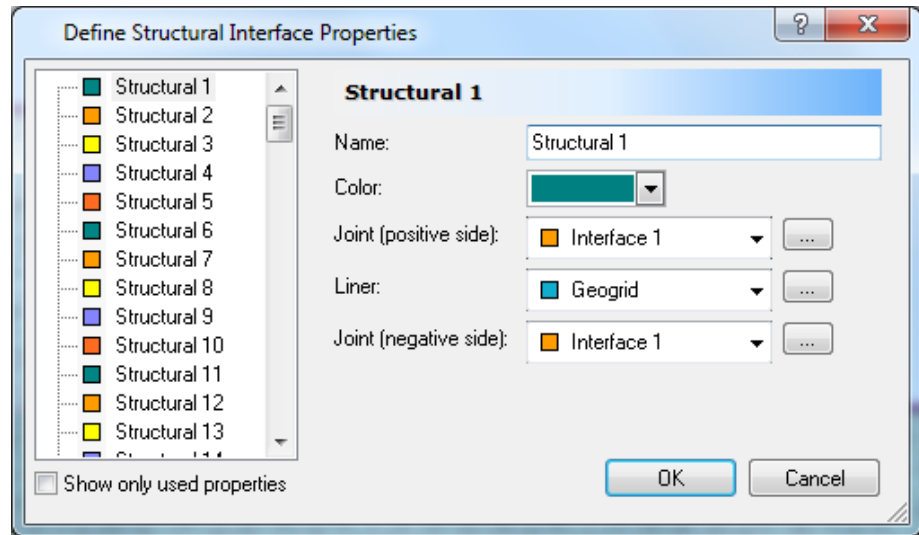
The Liner properties should already be assigned. Check by choosing Define Liners from the Properties menu. Ensure that for Liner 1 (Geogrid) the Liner Type is set to Geosynthetic, the Tensile modulus is 50000, the Material Type is Plastic and the Tensile peak and residual strengths are 60 and 0 respectively.

Now go to the Properties Menu and select Define Joints. For Joint 1, change the name to Interface 1. Set the Slip Criterion to Mohr-Coulomb and change the friction angle to 30 degrees. Leave all other parameters at their default values. The window should look like this:



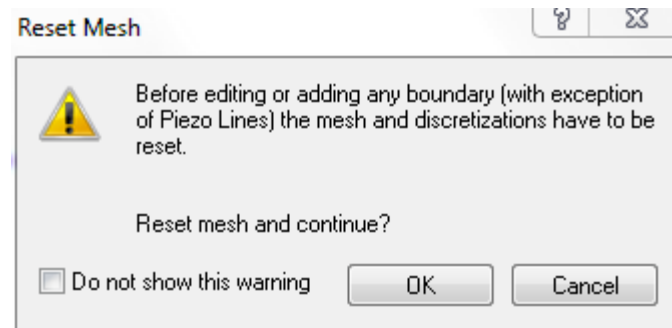
Click OK to close the window.

To assign properties to the Structural Interface, choose Define Structural Interface from the Properties menu. You can see how the Structural Interface is composed of a Liner sandwiched between two joints. You can specify different combinations of Liner and Joints to make up a Structural Interface. You can also change the properties of the liner or joints from this window by clicking on the ellipses (...) next to the joint or liner name. We will use the existing values so click OK to close the window.

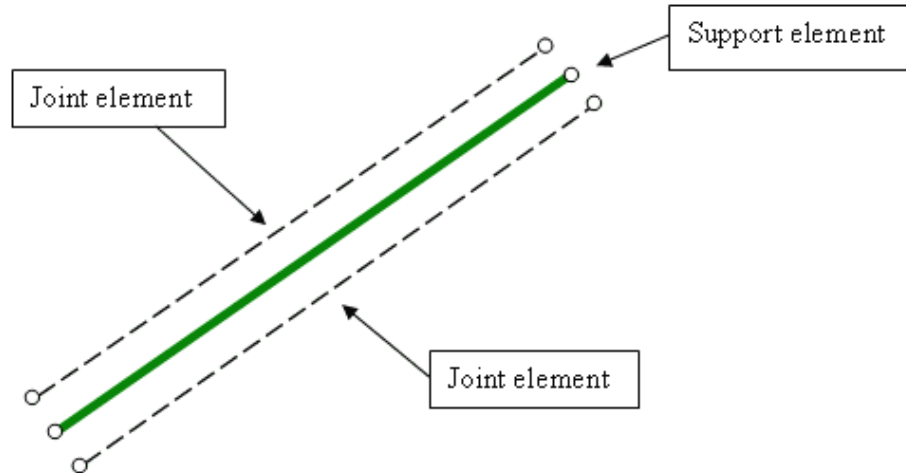


Add Support

Now we must add the Structural Interface to the model. Choose Add Structural Interface from the Boundaries Menu. You will be asked if you wish to reset the mesh as shown.



The reason that *Phase2* needs to reset the mesh is that the structural interface is composed of multiple nodes that initially overlap but may separate as slip occurs on the joints. The following schematic diagram illustrates the point.

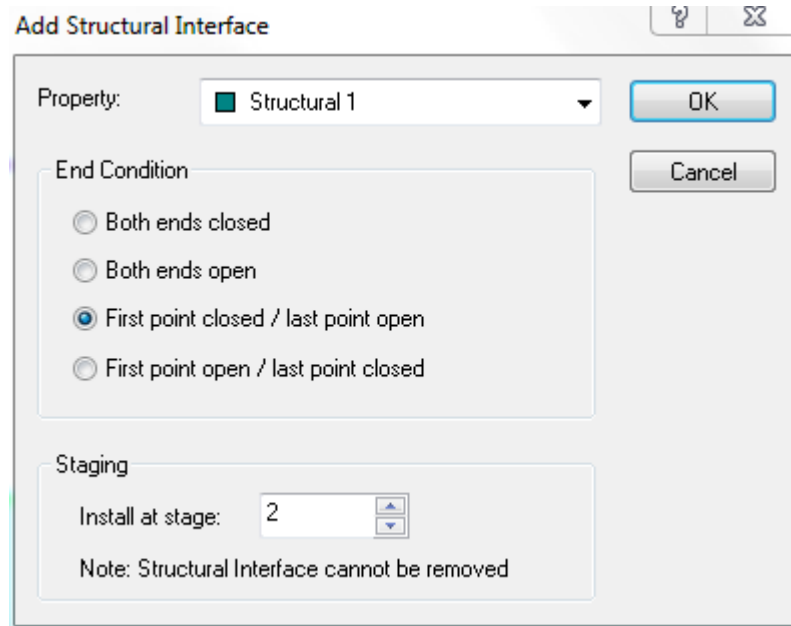


Click OK to reset the mesh.

You will now see the Add Structural Interface dialog. Choose Structural 1 for the Property. Now select First point closed / last point open for the end condition.

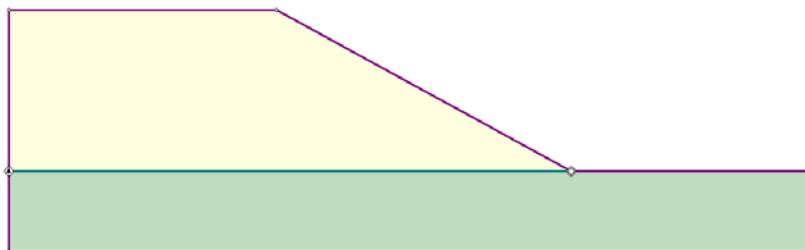
If a Joint end is *Closed*, this means that the end of the Joint boundary is represented by only ONE node in the finite element mesh, and therefore relative movement (sliding or opening) cannot occur at the joint end. If a Joint end is *Open*, this means that the end of the Joint boundary is represented by TWO nodes in the finite element mesh, which can move with respect to each other. In our model, one end of the joint will terminate at a free surface (the toe of the embankment) so this end should be set to Open. The end of the structural interface within the embankment will be defined as Closed.

We only want the geogrid to be installed at Stage 2 of the simulation. Therefore set the **Install at stage** option to 2. The dialog should look like this.



TIP: After you have installed the structural interface, you can still change the stage at which it is installed. To do this, view the desired stage and then right click on the Structural Interface and select **Install At This Stage**.

Click OK to close the dialog and begin selection of the boundary points. Click on the two end points of the existing material boundary (coordinates 0,3 and 21,3). Be sure to click the left point first to ensure that this is the closed end of the Structural Interface. Right click and choose **Done** or hit **Enter** to finish. The Structural Interface will now appear as a green line with a circled triangle indicating the closed end and an open circle representing the open end as shown.



HINT: You can also add the Structural Interface by using the *Phase2* **Convert Boundary** option to convert a Material boundary to a Structural Interface boundary. This is performed by selecting **Convert Boundary** from the **Edit** sub-menu of the **Boundaries** menu. See the *Phase2* Help for more information.

NOTE: in the *Phase2* Interpret program, the 3 components of the Structural Interface (joint/liner/joint) will be expanded and drawn with thick line segments for easy viewing.

You have now finished installing the geogrid.

Mesh

The model was unmeshed in order to add the Structural Interface. However this does not affect the mesh parameters. You therefore do not need to re-enter the mesh parameters. To remesh the model simply choose **Discretize and Mesh** from the Mesh menu.

You have now completed building the model. Save the model using the Save option in the File menu.

Compute

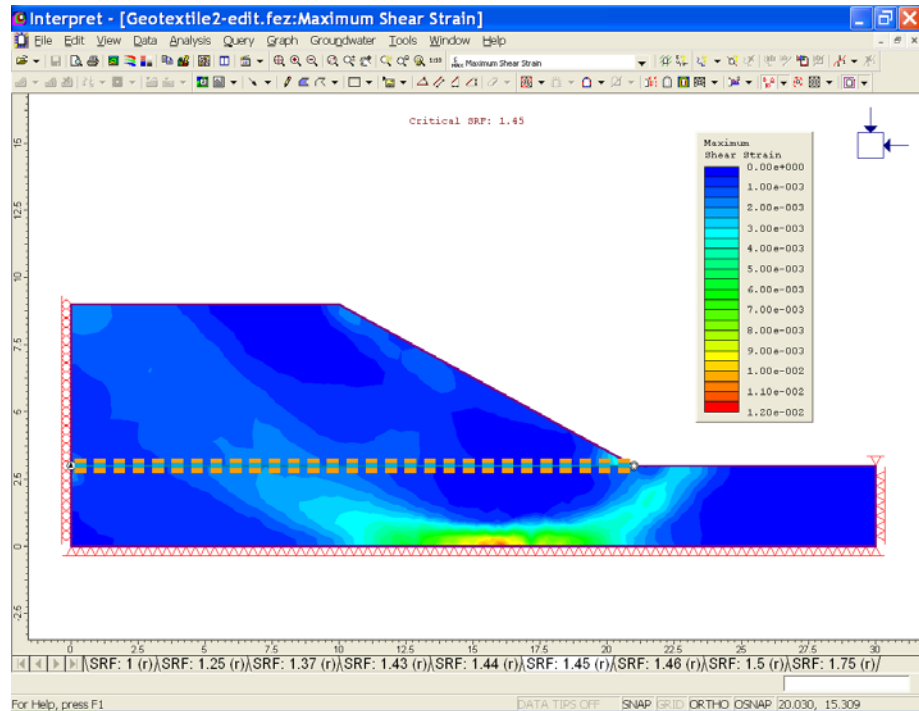
Run the model using the Compute option in the Analysis menu. The analysis should take a few minutes to run.

Once the model has finished computing (Compute dialog closes), select the Interpret option in the Analysis menu to view the results.

Interpret

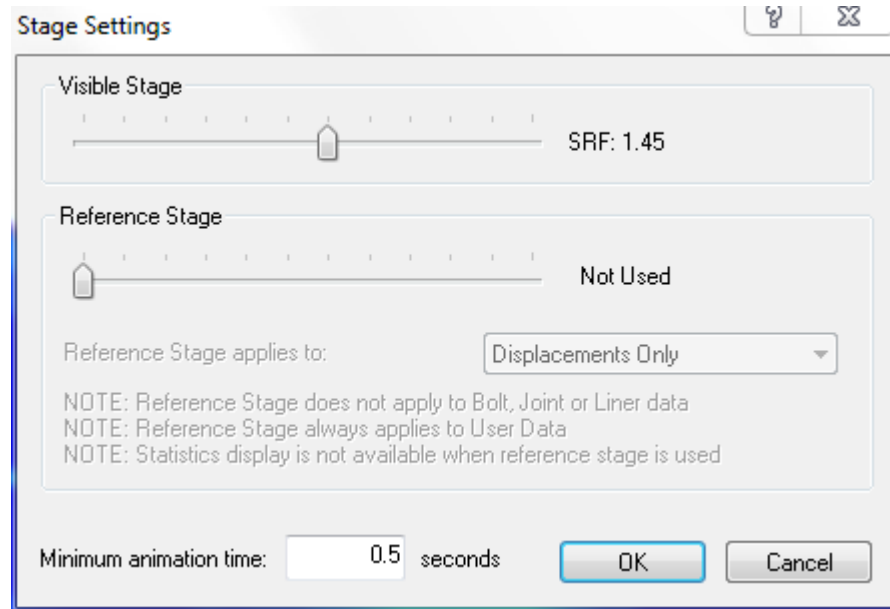
After you select the Interpret option, the Interpret program starts and reads the results of the analysis. The following screen is displayed showing the Maximum Shear Strain contours at the critical Strength Reduction Factor (SRF = 1.44).

TIP: Click and drag the legend to the right side of the screen so as not to cover up the model.



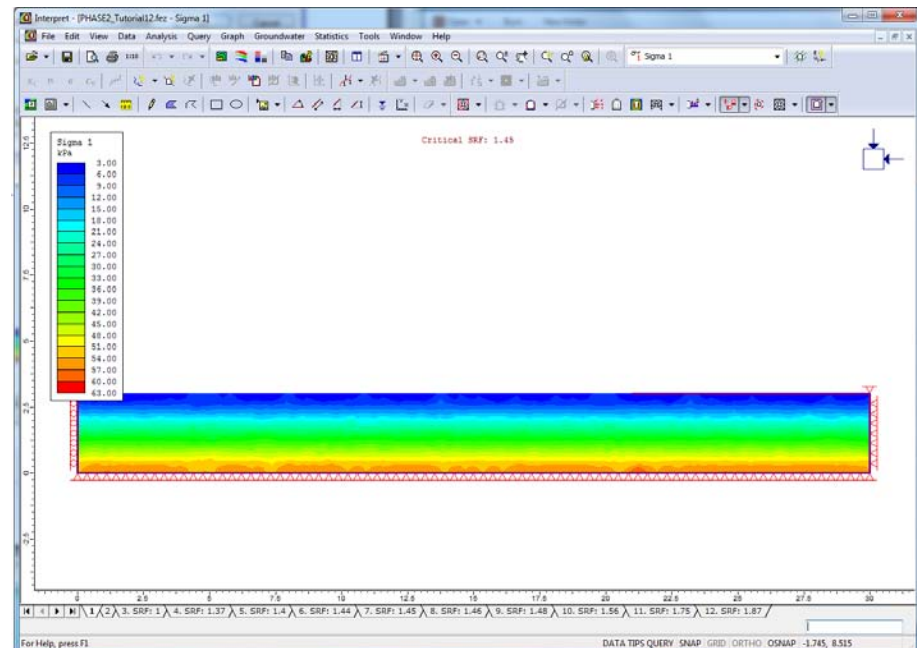
Before looking at the Shear Strength Reduction analysis, you may want to check the effect of the staging. You will see that the stress analysis results of Stage 1 and Stage 2 are not available to look at. Only the data for the different SRF values are available. To view the results from the different stages:

1. Select the Stage Settings option in the Data menu.
2. Move the reference stage slider all the way to the left so that it reads "Not Used".
3. Press the OK button to exit the dialog.



Stage 1 and Stage 2 tabs now exist along with the SRF tabs at the bottom of the view. You can now see the results of Stage 1 by selecting the Stage 1 tab.

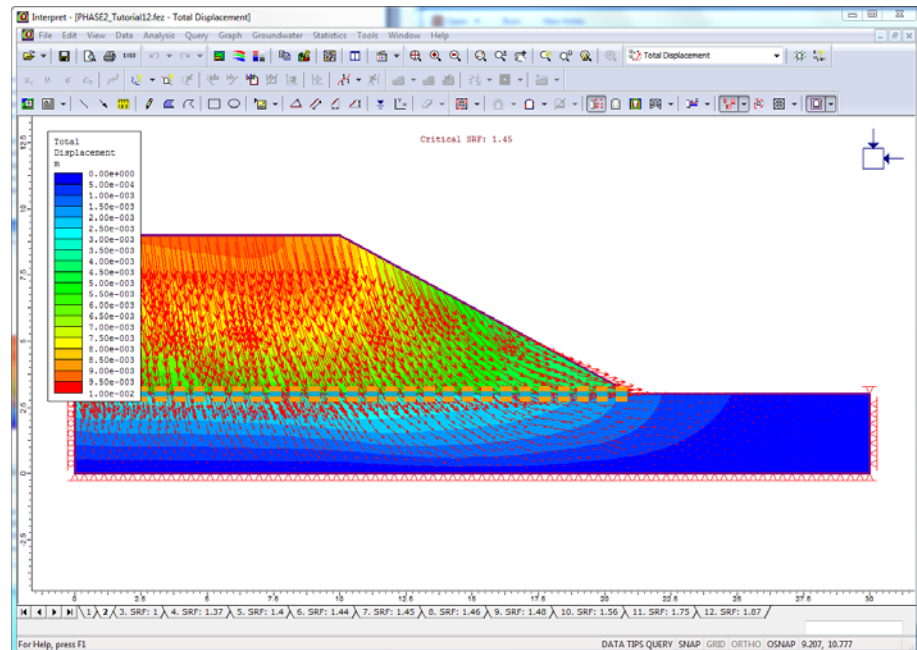
After selecting the Stage 1 tab, plot contours of maximum stress by choosing Sigma 1 from the drop down menu on the tool bar. Your model should now look like this.



You can see that the stress increases with depth in the clay layer due to gravitational load as you would expect. This stress is mostly due to the initial element loading. If you plot displacements, you will see that virtually no displacement has occurred.



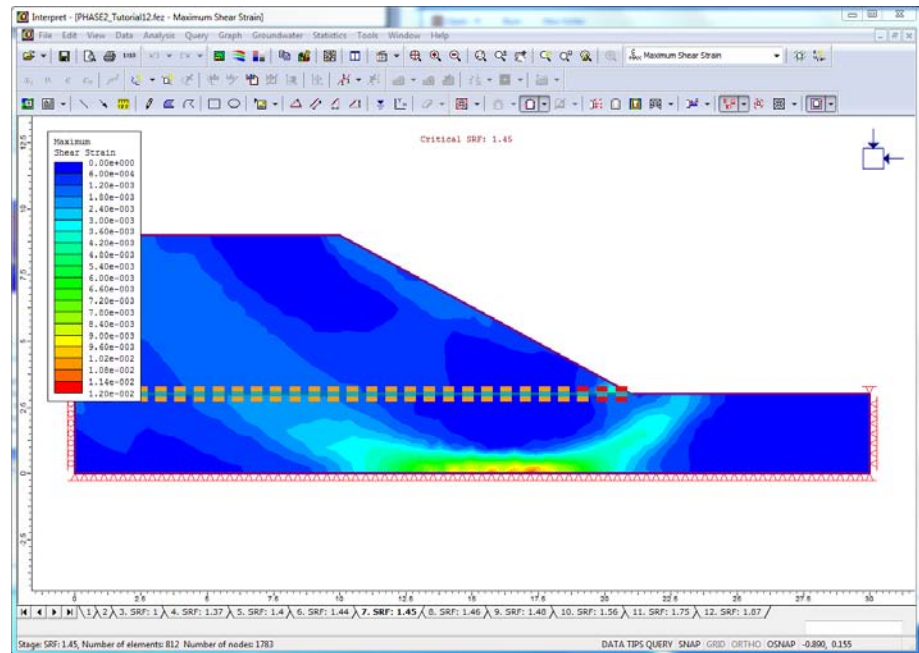
Click on the Stage 2 tab, and you can see larger stresses under the sand embankment. Change the contours to plot displacements by choosing Total Displacements from the drop down menu on the tool bar. Show the deformation vectors by selecting the Display Deformation Vectors button. Clearly a large amount of vertical displacement occurred in the sand and the clay as the soil layers compacted under the weight of the sand.



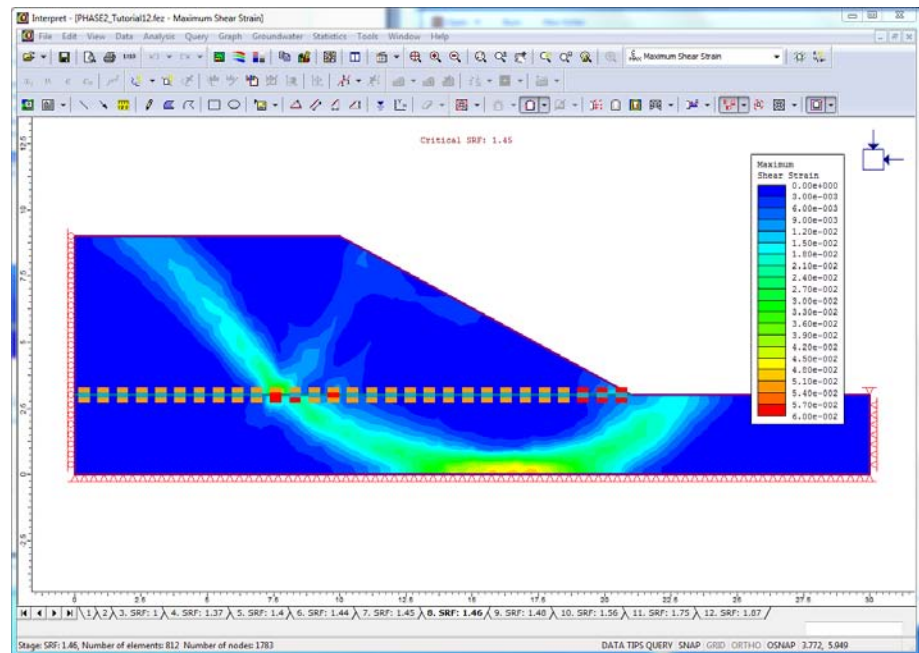
Click on the SRF=1 tab and you will see the same results. This is because SRF: 1 means that no strength reduction has been applied so these results are the same as the Stage 2 results. If you wish to look at deformations caused by strength reduction, rather than by settlement, you must go back to the Stage Settings item in the Data menu and set the reference stage back to SRF 1.



Now go back to the plot of SRF: 1.44 and change the contours back to Maximum Shear Strain. Turn off the displacement vectors. Display the yielded elements in the geogrid by clicking the Display Yielded Liners button. You will see that joint elements at the toe of the embankment have failed (slipped). Slippage has occurred on both sides of the geogrid, i.e. between the liner and the sand, and also between the liner and the clay. If you go back to the plot for SRF: 1 you will see the same slippage even before any shear strength reduction. This shows that the weight of the sand material has caused slip along the geogrid-material interfaces but that this slip is not responsible for the failure of the slope.



Now change the SRF to 1.45 by clicking on the SRF: 1.45 tab at the bottom of the window. Observe that one element in the geogrid itself has failed and that large shear strains accompany this failure. Clearly the tensile failure of the geogrid has resulted in unstable sliding of the slope (lack of convergence in the model).



Click through the other SRF plots and you will see further failure in the geogrid and the evolution of two localized shear bands as in the model from Tutorial 11.